

# Floating Point Arithmetic

## Fixed point Representation:

In our daily lives, we typically use fixed-point representation.

$$X = \pm (d_1 d_2 d_3 \dots d_{k-1} \cdot d_k \dots d_n)_\beta$$

$$\text{where } d_1 \dots d_n \in \{0, 1, \dots, \beta-1\}$$

### Example:

$$X = + (11.1)_2$$

$$X = - (21.5)_{10}$$

### Evaluating Fixed point numbers in base 10:

$$\left(\overset{1}{1}\overset{0}{1}\overset{-1}{.}1\right)_2 = 1 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1}$$

$$= 2 + 1 + \frac{1}{2}$$

$$= 3 + \frac{1}{2}$$

$$= \frac{7}{2}$$

$$= (3.5)_{10}$$

## Floating Point Representation

Floating point numbers are the subset of all real numbers  $F \subset \mathbb{R}$ . In computers, floating-point representation allows the system to handle a wide range of numbers, from tiny decimals to huge values, by breaking them down (organized) into a format that the computer can easily store and process.

$$F = \pm (0.d_1 d_2 d_3 \dots d_n) \beta \times \beta^e$$

Significand / Fraction / mantissa      Base      exponent

where,  $\beta, d_i, e \in \mathbb{Z} \rightarrow \text{integers}$

$$0 \leq d_i \leq \beta - 1$$

$$e_{\min} \leq e \leq e_{\max}$$

## Examples: (Magic of Math)

[We have convert a given number into floating point representation]

$$\begin{aligned} * & \pm (110.111)_2 \times 2^0 \\ &= \pm (11.0111)_2 \times 2^1 \\ &= \pm (1.10111)_2 \times 2^2 \\ &= \pm (0.110111)_2 \times 2^3 \end{aligned}$$

$$\begin{aligned} * & (1594.23)_{10} \times 10^0 \\ &= (159.423)_{10} \times 10^1 \\ &= (15.9423)_{10} \times 10^2 \\ &= (1.59423)_{10} \times 10^3 \\ &= (0.159423)_{10} \times 10^4 \end{aligned}$$

## Conventions

### Convention 1:

$$\pm (0.\underset{\substack{\downarrow \\ 1}}{d_1} d_2 d_3 \dots d_m) \beta \times \beta^e$$

\*\*\* Here  $d_1 = 1$  and  $d_1 \neq \text{any other value (always)}$  \*\*\*

$d_1$  always have to be 1.

Example: Given that,  $\beta = 2$ ,  $m = 3$ ,  $e \in [-1, 2]$   
 $e_{\min}$   $e_{\max}$

Highest possible Floating point number using convention:  
is  $(0.111)_2 \times 2^2$

### Normalized Form : (Convention 2)

$$\pm (1.d_1 d_2 d_3 \dots d_m)_\beta \times \beta^e$$

\* Here  $d_1$  can be 1 or any other value.

Example: Given that,  $\beta = 2$ ,  $m = 3$ ,  $e \in [-1, 2]$   
 $\downarrow$   $\downarrow$   
 $e_{\min}$   $e_{\max}$

Highest possible floating point number using Convention 2/  
Normalized Form is  $(1.111)_2 \times 2^2$ .

### Denormalized Form : (Convention 3)

$$\pm (0.1 d_1 d_2 d_3 \dots d_m)_\beta \times \beta^e$$

\* Here  $d_1$  can be 1 or any other value.

Example: Given that,  $\beta = 2$ ,  $m = 3$ ,  $e \in [-1, 2]$

Highest possible floating point number using  
Convention 3/denormalized form is  $(0.1111)_2 \times 2^2$ .

Total Combination (How to find total Combination)  
How many floating numbers can be possible

1. Given that,  $\beta = 2$ ,  $m = 3$ ,  $e = [-1, 2]$   
 $\downarrow \quad \downarrow$   
 $e_{\min} \quad e_{\max}$

Using convention 1: [where  $d_1 = 1$  (always)]

For  $e = -1$

$$(0.1 \underline{d_2} \underline{d_3})_2 \times 2^{-1}$$

|   |   |
|---|---|
| 0 | 0 |
| 0 | 1 |
| 1 | 0 |
| 1 | 1 |

[For  $e = -1$ , Here 4 combinations possible]

$$2^n = 2^2 = 4$$

$e = [-1, 2]$  means ' $e$ ' can be  $-1, 0, 1, 2$ .

For  $e = 0$

$$(0.1 \underline{d_2} \underline{d_3})_2 \times 2^{-1}$$

|   |   |
|---|---|
| 0 | 0 |
| 0 | 1 |
| 1 | 0 |
| 1 | 1 |

[For  $e = 0$ , Here 4 combinations possible]

Same thing goes for

$$\boxed{e = 1}, \quad \boxed{e = 2}$$

4 combinations possible

4 combinations possible

$\therefore$  total possible combinations  $= 4 + 4 + 4 + 4 = 16$ .

2. Given that,  $\beta=2$ ,  $m=3$ ,  $e=[-1, 2]$

\* Using normalized form / Convention 2

$$\left(1. \overline{d_1} \overline{d_2} \overline{d_3}\right)_2 \times 2^{-1}$$

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |
| 1 | 1 | 1 |

[For  $e=-1$ , Here 8 combinations possible]

$$2^n = 2^3 = 8$$

Same thing goes for  $\underbrace{e=0}_{8 \text{ combination}}, \underbrace{e=1}_{8 \text{ combination}}, \underbrace{e=2}_{8 \text{ combination}}$

$$\therefore \text{Total possible Combinations} = 8 + 8 + 8 + 8 \\ = 32$$

3. Given that  $\beta = 2$ ,  $m = 3$ ,  $e = [-1, 2]$

\* Using normalized Form / Convention 3

$$(0.1 \frac{d_1}{2} \frac{d_2}{2} \frac{d_3}{2})_2 \times 2^{-1}$$

$$0 \ 0 \ 0$$

$$0 \ 0 \ 1$$

$$0 \ 1 \ 0$$

$$0 \ 1 \ 1$$

$$1 \ 0 \ 0$$

$$1 \ 0 \ 1$$

$$1 \ 1 \ 0$$

$$1 \ 1 \ 1$$

[For  $e = -1$ , Here 8 combinations possible]

$$\boxed{2^3 = 8}$$

Same thing goes for  $e = 0$ ,  $e = 1$ ,  $e = 2$

8 combinations    8 combinations    8 combinations

$$\begin{aligned} \therefore \text{Total possible combinations} &= 8 + 8 + 8 + 8 \\ &= 4 \times 8 \\ &= 32 \end{aligned}$$

## \* Practice Problem

[Find the smallest and largest values]

1. Given that  $\beta = 2$ ,  $m = 3$ ,  $e = [-1, 2]$ . Find the largest or maximum value or Highest Value.

[Using Convention 1]

$$\begin{aligned} \text{Highest value} &= + (0 \cdot d_1 d_2 d_3)_2 \times 2^e \\ &= + (0 \cdot \underline{1} \underline{d_2} \underline{d_3})_2 \times 2^e \\ &= + (0 \cdot \underline{1} \ 1 \ 1)_2 \times 2^{\textcircled{2}} \rightarrow l_{\max} \\ &\quad d_1 = 1 \end{aligned}$$

[As we follow convention 1 here]

$$= (7/2)_{10}$$

2. Given that  $\beta = 2$ ,  $m = 3$ ,  $e = [-1, 2]$ , Find the non-negative smallest or minimum value.

[Using Convention 1]

$$\begin{aligned} \text{Smallest value [non-negative]} &= + (0 \cdot \underline{1} \ 0 \ 0)_2 \times 2^{\textcircled{-1}} = \frac{1}{4} \\ &\quad d_1 = 1 \quad \downarrow \quad l_{\min} \end{aligned}$$

[as we follow convention 1]

3. Give that  $\beta = 2$ ,  $m = 3$ ,  $e = [-1, 2]$

Find the negative minimum or smallest value.

$$\begin{aligned} \text{Negative minimum or smallest value} &= - \text{Highest value} \\ &= - (0 \cdot 1 \ 1 \ 1)_2 \times 2^2 \\ &= - 7/2 \end{aligned}$$

\*\*\* Important \*\*\*

\* IF it mentions only the smallest number, you need to include both the smallest non-negative and negative value.

Note

## Binary to Decimal

$$\begin{aligned} 1. & \quad + (0. \overset{-1}{1} \overset{-2}{1} \overset{-3}{1})_2 \times 2^2 \\ &= (1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}) \times 2^2 \\ &= \left( \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} \right) \times 2^2 \\ &= \left( \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \right) \times 2^2 \\ &= \frac{4+2+1}{8} \times 4 \\ &= \left( \frac{7}{2} \right)_{10} = 7/2 \end{aligned}$$

$$\begin{aligned} 2. & \quad (0. \overset{-1}{1} \overset{-2}{0} \overset{-3}{0})_2 \times 2^{-1} \\ &= (1 \times 2^{-1} + 0 \times 2^{-2} + 0 \times 2^{-3}) \times \frac{1}{2} \\ &= \frac{1}{2} \times \frac{1}{2} \\ &= \frac{1}{4} \end{aligned}$$

## Practice Problem

Ex 1  $\beta = 2$ ,  $m = 3$ ,  $e = [-1, 2]$ . Find the highest value using Normalized Form.

$$\pm (1 \cdot d_1 d_2 d_3 - \dots - d_m)_{\beta} \times B^e$$

$$\text{Highest value} = + (1 \cdot 1 \cdot 1 \cdot 1)_2 \times 2^e$$

$$= (1 \cdot 1 \cdot 1 \cdot 1)_2 \times 2^2$$

$$= (1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}) \times 2^2$$

$$= \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}\right) \times 2^2$$

$$= \left(1 + \frac{4+2+1}{8}\right) \times 2^2$$

$$= \frac{15}{8} \times 2$$

$$= \left(\frac{15}{2}\right)_{10}$$

$$= (7.5)_{10}$$

Q1  $\beta = 2, m = 3, e = [-1, 2]$ . Find the highest value using denormalized Form.

$$\pm (0.1 d_1 d_2 d_3 \dots d_m)_{\beta} \times \beta^e$$

$$\text{Highest value} = (0.1111)_{\beta} \times 2^2$$

$$= (1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3} + 1 \times 2^{-4}) \times 2^2$$

$$= \left( \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} \right) \times 4$$

$$= \frac{8+4+2+1}{16} \times 4$$

$$= \frac{15}{4}$$

$$= \left( \frac{15}{4} \right)_{10}$$

$$= (3.75)_{10}$$

## Floating Point numbers are not equally spaced

Let  $\beta=2$ ,  $m=3$ ,  $e=[-1, 1]$

Forc convention 1

\* Smallest non-negative numbers combinations.

For  $e = -1$ ,

$$(0.100) \times 2^{-1} = \frac{1}{4}$$

$$(0.101) \times 2^{-1} = \frac{5}{16}$$

$$(0.110) \times 2^{-1} = \frac{3}{8}$$

$$(0.111) \times 2^{-1} = \frac{7}{16}$$

For  $e = 0$

$$(0.100) \times 2^0 = \frac{1}{2}$$

$$(0.101) \times 2^0 = \frac{5}{8}$$

$$(0.110) \times 2^0 = \frac{3}{4}$$

$$(0.111) \times 2^0 = \frac{7}{8}$$

For  $e = 1$ ,

$$(0.100) \times 2^1 = 1$$

$$(0.101) \times 2^1 = \frac{5}{4}$$

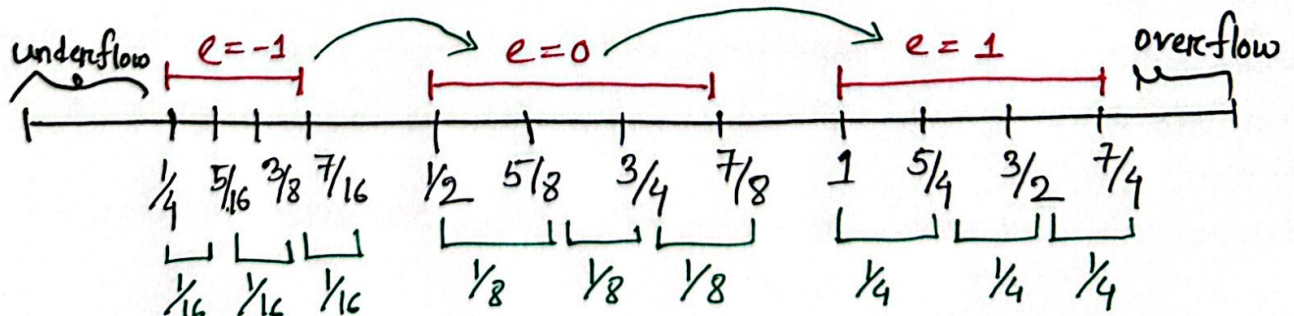
$$(0.110) \times 2^1 = \frac{3}{2}$$

$$(0.111) \times 2^1 = \frac{7}{4}$$

\*\*\* important \*\*\*

When exponent  $e$  is constant, numbers are equally spaced.

However, when exponent  $e$  changes, the interval or gap between numbers change.



## IEEE Standard for double precision (64 bit)

$$\beta = 2$$

$$\text{Fraction/mantissa} = 52 \text{ bits}$$

$$\text{Exponent} = 11 \text{ bits}$$

$$\text{Sign} = 1 \text{ bits}$$

Using Normalize Form

$$\pm (1.d_1 d_2 d_3 \dots d_{52})_2 \times 2^e$$

For  $e = 11$  bits, there will be total  $2^{11} = 2048$ .

$$\text{range of } e \quad \left[ \underset{\substack{\downarrow \\ e_{\min}}}{0}, \underset{\substack{\downarrow \\ e_{\max}}}{2047} \right]$$

$$\text{Largest possible number} = + (1.111 \dots 1)_2 \times 2^{2047}$$

$$\begin{aligned} \text{Smallest possible number (non-negative)} &= + (1.00 \dots 0)_2 \times 2^0 \\ &= 1 \end{aligned}$$

\* To get a small number, the exponent value must be small.

\* We obtain smaller values through exponent biasing.

[1 is not the enough small number. In this way FP can't store smaller values]

Formula:

$$\begin{aligned} & 2^{K-1} - 1 \\ &= 2^{11-1} - 1 \\ &= 2^{10} - 1 \\ &= 1024 - 1 = 1023 \end{aligned}$$

$$\therefore (1.d_1 d_2 \dots d_{52})_2 \times 2^{e - \frac{1023}{2}} \quad \text{Exponent biasing}$$

For denormalized form

$$(0.1 d_1 d_2 \dots d_{52})_2 \times 2^{e-1022}$$

$$e-1022 \in [-1022, 1025]$$

$$\begin{aligned} \text{Largest possible number} &= (0.11 \dots 1)_2 \times 2^{1025} \\ &\approx \infty \end{aligned}$$

$$\begin{aligned} \text{Smallest possible number} &= (0.10 \dots 0)_2 \times 2^{-1022} \\ &\approx 0 \end{aligned}$$



\* Reduce the range for largest possible number so that we can get the smaller values.

$$2^{\frac{1025}{2}} \rightarrow \pm \infty \quad [\text{highest power is used to store } \infty]$$

$$2^{-\frac{1022}{2}} \rightarrow \pm 0 \quad [\text{used to store } 0]$$

$$\therefore \text{Highest value} = (0.111\dots 1)_2 \times 2^{1024} \quad [\text{if we put } 1025 \text{ it will denote } \infty]$$

$$\approx 1.798 \times 10^{308}$$

$$\therefore \text{Smallest value} = (0.100\dots 0)_2 \times 2^{-1022}$$

$$\approx 2.225 \times 10^{-308}$$

